Vipan Kumar Garg

Age - 66 | Height - 5’ 11’ | Weight - 74 Kg / 165 lbs

Patient History

- Asthma since 20 yrs (takes foracort 200 inhaler + Montair LC regularly)
- Hypertension/BP (since last 5 years) - takes medicine daily (Telmisartan (40mg)
- Diagnosed as pre-diabetic in Jan 2022 (Hb1AC - 6)
- Had thyroid since last 5 years but took medicines and it is under control now
- Has prostate and takes Alfuzosin (10mg) daily for that

Multiple Myeloma History

- **April 2022**: Started complaining of back pain in April 2022 (also had covid in the same month), blood tests showed high ESR - 90
- **May 2022**: Patient also fell one morning in the bathroom in May 2022, got unconscious (probably due to hypertension) but nothing was detected (Chest X rays done but nothing showed up)
- **June 2022:** Bone Marrow / Serum Electrophoresis done in June 2022 and Multiple Myeloma confirmed (Patient’s mother also suffered from Multiple Myeloma, Patient’s mothers brother also suffered from Multiple Myeloma)
    - Multiple Myeloma confirmed due to Bone Lesions found in spine and some ribs (C-ve, A -ve, R-ve, B+ve in CARB diagnosis). D1 and D6 have collapsed (compression fractures) (MRI attached below)

  FISH report: Abnormal FISH results: 3-4 copies each of 1q21(CKS1B locus) and 1p32(CDKN2C locus) seen in 10% of cells suggestive of trisomy/tetrasomy of chromosome 1. Normal FISH results: No evidence of 1p32(CDKN2C) deletion,t(4;14)(p16;q32) FGRF3::IGH, t(14;16)(q32;q23)IGH::MAF, t(14;20)(q32;q12)IGH::MAFB translocations,11q13(CCND1) rearrangement and17p13(TP53 deletion). Note: FISH done on non-sorted sample.
- **July 2022 - Oct 2022:** Treatment Cycle of DaraVCD: Chemotherapy started on 20th June 2022 and has undergone 5 cycles. 12 Dara infusions + 16 weekly Bortezomib.
    - Weekly dose of **Bortezomib + Dexamethasone** (40mg) + **Cyclophosphamide** + **Daratumumab** (400mg x 3) . Patient started with Lenalidomide but was replaced with Cyclophosphamide as platelet count was going down after the patient was given Lenalidomide for 4 weeks.
    - Weekly dose of Zoledronic acid for first 4 months
- **Dec 2022**: MRD -ve - tested NIL and no change in bone lesions in PET scan. Hence Bone Marrow Transplant was ***canceled*** since he achieved complete remission (CR).

- **Jan 2023 - Aug 2023**: Maintenance therapy of Biweekly **Bortezomib 1.25mg** (given subcutaneously). Quarterly Zoledronic acid.
- **Aug 2023**: Repeat PET scan and MRD test done. MRD tested negative and PET scan showed no change. Patient traveled to the US and spent 3 months with a completely normal lifestyle.
- **Dec 2023 - Jan 2024**: Myeloma follow up showed an increase in Kappa FLC to 50 that raised suspicion of relapse in Dec 23 test. Jan 2024 retested with a different lab - Kappa FLC increased again to 98.
- **February 2024**:
    - **Bone Marrow Test (Report attached):**

No of viable events (FSC vs SSC): 90%

CD38+CD138+ Events: 0.15%

Report Summary: Flowcytometric immunophenotyping done on bone marrow aspirate show plasma cells with immunophenotype as follows:

1. NPC (Normal Plasma Cells): 0.04%; CD45DIMCD19+CD27+CD20dimCD56-CD117- distribution (kappa: lambda = 1.7:1)

APC (Aberrant Plasma Cells): 0.1%; Kappa restricted plasma cells displaying two disti
subpopulation I: CD19+CD27+CD56+
subpopulation II: CD19-CD27-CD56+

clusion: Sample is positive for MRD (0.1%)

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    - **PET Scan Test (Report attached):**
        - 1. Metabolically active lytic lesions involving few skeletal sites as described (pathological fractures involving right 6th & left 8th ribs, collapsed D6 vertebra) - residual / recurrent neoplastic disease involvement.
        - 2. Non-metabolic lytic lesions in multiple other skeletal sites as described above; D1 & D2 vertebral collapse with no significant metabolic activity.
        - 3. Also noted:
            - (i) Metabolically active multiple irregular fibronodules, few arranged in tree-in-bud pattern with adjacent ground glass haziness in the left lung with multiple mediastinal & right supraclavicular lymph nodes - favors active infective/inflammatory etiology.
            - (ii) Hepatomegaly.
            - (iii) Prostatomegaly.
        - 4. No definite scan evidence of abnormal metabolically active disease in the rest of the body in this study.

- **March 2024**: Doctors recommended to get CAR-T treatment asap if possible as the best line of treatment (since the patient is not ASCT eligible)

Kappa/Lambda Free Light Chains

	MM detected	DARA-VCD - 4 cycles done	MRD -ve				
	May 2022	July 2022	Feb 2023	May 2023	Aug 2023	Dec 2023	Jan 2024
Serum Kappa FLC	857	16.97	8.08	16.65	23.08	56.82	92.94
Serum Lambda FLC	17.8	11.7	4.19	13.74	12.71	17.28	18.35
Kapp / Lambda Ratio	48.3	1.45	1.93	1.21	1.82	3.29	5.065

PROTEIN ELECTROPHORESIS, SERUM

	MM detected	DARA-VCD - 4 cycles done	MRD -ve				
	May 2022	July 2022	Feb 2023	May 2023	Aug 2023	Dec 2023	Jan 2024
Protein, Total	8.4	5.8	6.5	6.9	6.7	8.0	7.34
Albumin	4.1	1.949	2.254	1.907	4.44	1.69	4.46
Alpha 1 globulin	0.3	0.257	0.17	0.142	0.18	0.129	0.27
Alpha 2 globulin	0.8	0.606	0.536	0.477	0.69	0.425	0.7
Beta 1 globulin	0.5	0.617	0.688	0.462	0.75	0.528	0.5
Beta 2 globulin	0.3						0.31
Gamma globulin	2.4	0.512	0.295	0.468	0.64	0.677	1.1
A : G Ratio	0.95	0.98	1.33	1.23	1.96	0.96	1.55
M Spike g/d	2.1	0.22	0.04	No M spike	Very faint band	0.24	0.2

Blood Tests

Haemogram	Dec 2022	Jan 2023	Mar 2023	Apr 2023	May 2023	Aug 2023	Dec 2023
Haemoglobin	12.3	11.5	11.7	12.1	11.7	11	10.4
Packed Cell. Volume	36.1	34.3	35.4	36.6	36.7	34.5	32.2
Total Leucocyte Count (TLC)	7.7	4.6	6.4	5.2	5	4.8	5.8
RBC count	3.77	3.9	3.68	3.91	3.96	3.96	3.96
MCV	95.7	98	96.2	93.6	92.6	87	81.2
MCH	32.7	32.8	31.8	30.9	29.5	27.7	26.1
MCHC Calcuiated	34.2	33.4	33.1	33	31.8	31.8	32.2
Platelet Count	14	150	150	124	150	143	199
MPV	10.1	9.8	11.4	11.2	9.9	10.6	8.8
ROW Calculated	16	15.6	13.6	13.6	14.4	15.4	15.8
Neutrophils	81	76	56	59.4	63.7	60.2	61.4
Lymphocytes	8.1	10.2	17	22.8	21.6	26.2	21.7
Monocytes	10.3	12.7	8	12.5	12.2	10.5	14.4
Eosinophils	0.5	1	19	5.2	2.2	2.6	2
Basophils	0.1	0.1	0	0.1	0.3	0.5	0.5
Absolute Leukocyte Count calculated from TLC & DLC							
Absolute Neutrophil Count	6.24	3.5	3.58	3.09	3.19	2.89	3.56
Absolute Lymphocyte Count	0.6	0.5	1.1	1.2	1.1	1.3	1.3
Absolute Monocyte Count	0.79	0.58	0.51	0.65	0.61	0.5	0.84
Absolute Eosinophil Count	0.04	0.05	1.22	0.27	0.11	0.12	0.12
Absolute Basophil Count	0.01			0.01	0.02	0.02	0.03
ESR (Westergren)	6	45	80	25	22	22	33
HbA1c (Glycated/ Glycosylated Hemoglobin) Test							
Glucose (Fasting)	81	87	82	91	85	95	92
Glycosylated Haemoglobin(Hb AIC)	6	6.1	5.5	5.6	5.9	5.8	6.1
Glycosylated HaernOg10bin(Hb AIC) IFCC	42.1	43.16	36.6	37.69	40.97	39.88	43.16
Average Glucose Value For the Last 3 Months	126	128.37	111.15	114.02	122.63	119.76	128.37
Average Glucose Value For the Last 3 Months IFCC	6.95	7.11	6.16	6.31	6.79	6.63	7.11

Thyroid Profile*,Serum							
Free Triodothyronine (FT3) CLIA	2.43	3.2	2.55	2.95	3.07	2.86	3.44
Free Thyroxine (FT4) CLIA	0.73	0.72	0.58	0.78	0.67	0.67	0.83
Thyroid Stimulating Hormone CLIA	3.94	3.066	4.113	1.66	2.055	2.887	3.784
Kidney Function Test (KFT) Profile with Calcium, Uric Acid							
Urea Urease, UV	22	22	24	29	29	31	30
Blood Urea Nitrogen Urease, UV	10.3	10.28	11.21	13.55	13.55	14.49	14.02
Creatinine Akaline picrate knetiC	0.98	0.94	1.15	1.19	1.19	1.3	1.32
eGFR	76.6	80.35	63.64	61.16	61.15	60.26	59.06
Bun/Creatinine Ratio Calcudated	10.5	10.94	9.75	11.39	11.39	11.15	10.62
Uric Acid Uricase, Colorimetric	4.3	4.5	4.6	4.9	4	6	5.8
Calcium (Total)	9.42	9.11	9.29	10.04		9.46	9.69
Sodium ISE indirect	134	136	141	141	139	138	137
Potassium	4.81	5.05	5.28	4.95	4.03	4.75	4.83
Chloride ISE intf,rect	97	102	106	106	106	105	103
Bicarbonate	31.2	31	29	27.4	27.4	28.1	28.9
Liver Function Test (LFT)							
Total Protein Biuret	6.34	5.66	6.52	6.52	6.52	6.97	7.07
Albumin	4.1	3.8	4.1	4.3	4.2	4.3	4.2
Globulin Calculated	2.2	1.8	2.4	2.2	2.3	2.5	2.8
A G, ratio	1.8	2.1	1.7	1.9	1.9	1.8	1.5
Bilirubin (Total) tre	0.45	0.39	0.42	0.48	0.41	0.34	0.33
Bilirubin (Direct) Diazobzation	0.11	0.05	0.08	0.09	0.07	0.03	0.05
Bilirubin (Indirect) Calculated	0.34	0.34	0.34	0.39	0.34	0.31	0.28
SGOT• Aspartate	18	17	25	24	25	25	19
Transaminase (AST) UV w/hout P5P SGPT- Alanine Transaminase (ALT) UV without P5P	15	14	27	14	14	17	11
AST/ALT Ratio	1.2	1.21	0.93	1.71	1.79	1.47	1.73
Alkaline Phosphatase P•NPP. Mlp El-ner	106	78	117	85	74	83	84
GGTP (Gamma GT), Serum	22	21	15	13	13	15	14

Lipid Profile,Serum							
Cholesterol Cholesterol oxidase. esterase. peroxidese	174	207	190	218	212	263	214
HDL Cholesterol Direct measure, immunoinhibieon	65	62	49	52	57	58	51
LDL Cholesterol	100	126	112	133	131	174	149
Triglyceride Enzymatic. end point	101	163	147	148	218	203	137
VLDL Cholesterol Calculated	20.2	326	29.4	29.6	43.6	40.6	27.4
Total Cholesterol/HDL Ratio Calculated	2.7	3.3	3.9	4.2	3.7	4.5	4.2
Non-HDL CtDlester01 Calculated	109	145	141	166	155	205	163
HDL.,LDL	0.65	0.49	0.44	0.39	0.44	0.33	0.34